EXPERIMENT- 10

Python Code:

```python
matrix = [
    ['M', 'F', 'H', 'I', 'K'],
    ['U', 'N', 'O', 'P', 'Q'],
    ['Z', 'V', 'W', 'X', 'Y'],
    ['E', 'L', 'A', 'R', 'G'],
    ['D', 'S', 'T', 'B', 'C']
]
def get_pos(c):
    for i in range(5):
        for j in range(5):
            if matrix[i][j] == c:
                return i, j
    return -1, -1

plaintext = input("Enter plaintext: ").upper().replace("J", "I")
filtered_text = ""
for char in plaintext:
    if char.isalpha():
        filtered_text += char

prepared_text = ""
i = 0
while i < len(filtered_text):
    a = filtered_text[i]
    if i + 1 < len(filtered_text):
        b = filtered_text[i+1]
        if a == b:
            prepared_text += a + "X"
            i += 1
        else:
            prepared_text += a + b
            i += 2
    else:
        prepared_text += a + "X"
        i += 1
encrypted_text = ""
for i in range(0, len(prepared_text), 2):
    r1, c1 = get_pos(prepared_text[i])
    r2, c2 = get_pos(prepared_text[i+1])
    if r1 == r2:
        encrypted_text += matrix[r1][(c1+1)%5] + matrix[r2][(c2+1)%5]
    elif c1 == c2:
        encrypted_text += matrix[(r1+1)%5][c1] + matrix[(r2+1)%5][c2]
    else:
        encrypted_text += matrix[r1][c2] + matrix[r2][c1]
print("Encrypted Text:", encrypted_text)
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\10.py"
Enter plaintext: Must see you over Cadogan West. Coming at once.
Encrypted Text: UZTBDLGZPNNWLGTGTUEROVLDBDUHFPERHWQSRZ
PS D:\College\Crypto\Experiments>
```